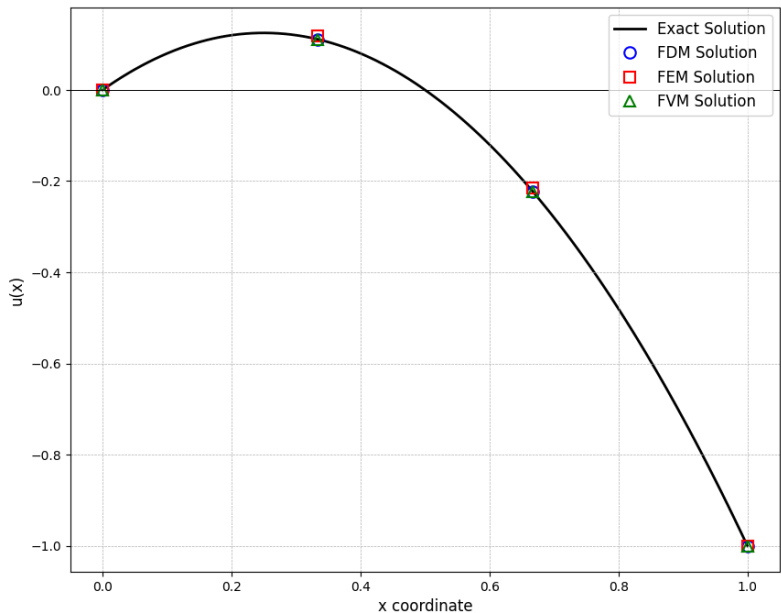


FINAL COMPARISON BETWEEN FDM, FEM & FVM

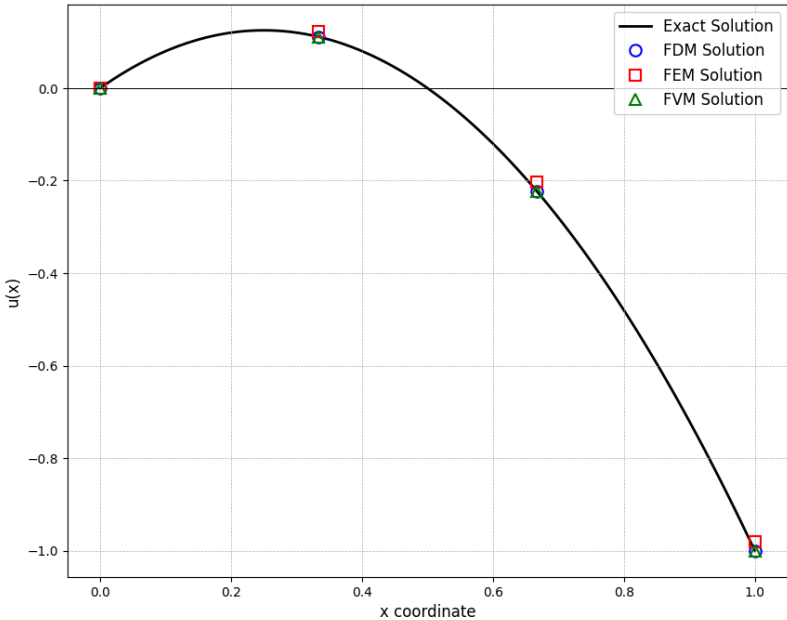
1. Dirichlet Boundary Conditions				
	Exact Solution	FDM	FEM	FVM
U1 ($x=0$)	0	0	0	0
U2 ($x=\frac{1}{3}$)	0.1111	0.1111	0.1181	0.1077
U3 ($x=\frac{2}{3}$)	-0.2222	-0.2222	-0.2153	-0.2256
U4 ($x=1$)	-1	-1	-1	-1

2. Neumann Boundary Condition				
	Exact Solution	FDM	FEM	FVM
U1 ($x=0$)	0	0	0	0
U2 ($x=\frac{1}{3}$)	0.1111	0.1111	0.1231	0.1062
U3 ($x=\frac{2}{3}$)	-0.2222	-0.2222	-0.2039	-0.2291
U4 ($x=1$)	-1	-1	-0.9798	-1.0062

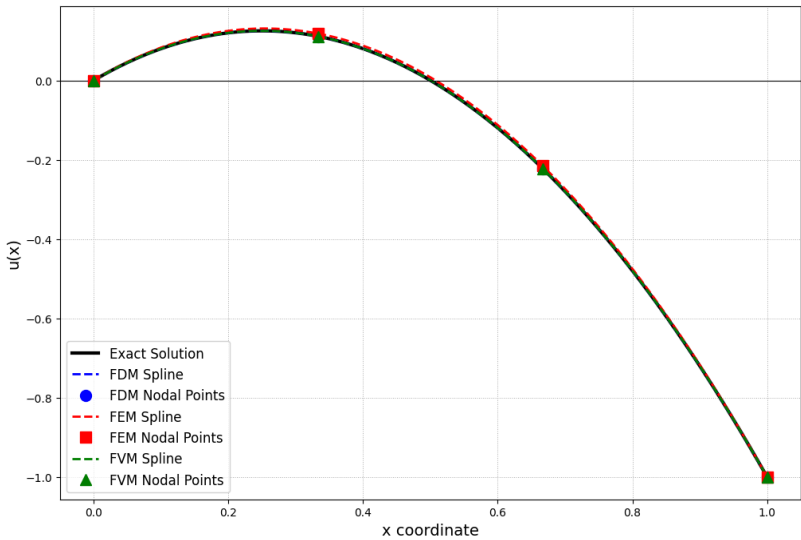
Comparison of Numerical Methods
(Dirichlet Conditions)



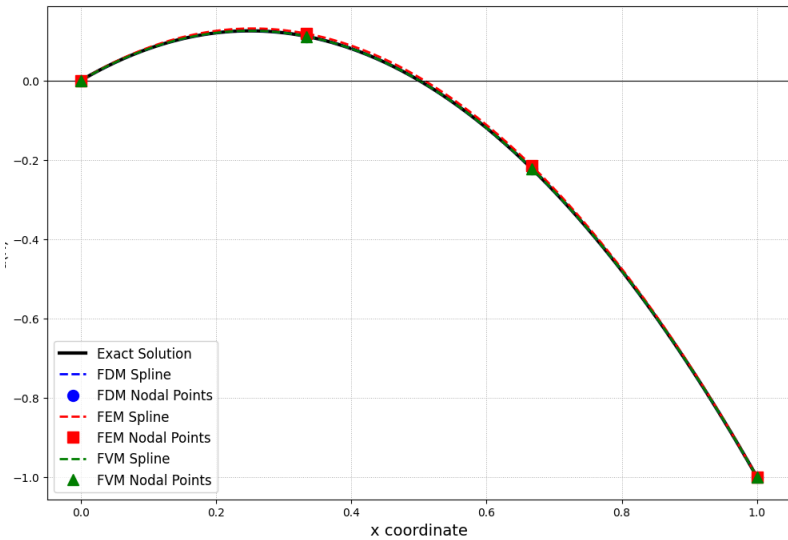
Comparison of Numerical Methods
(Neumann Conditions)



Spline Comparison of Numerical Methods
(Dirichlet Conditions)



Spline Comparison of Numerical Methods
(Dirichlet Conditions)



Key Observations

- **Finite Difference Method (FDM) Performance:** The most significant finding is that the **FDM produced results identical to the exact solution** for all nodes under both Dirichlet and Neumann boundary conditions. This suggests that for this particular 1D second-order problem with a uniform grid, the truncation errors of the central difference scheme either canceled out perfectly or were zero.
- **FEM and FVM Performance:** Both the **Finite Element Method (FEM)** and the **Finite Volume Method (FVM)** provided good approximations of the exact solution, but both introduced small **discretization errors**.
- **Boundary Condition Handling:**
 - All three methods perfectly enforced the **Dirichlet boundary conditions**, as the nodal values were set directly.

Overall Conclusion

For this specific assignment, the **Finite Difference Method was demonstrably the most accurate method**, yielding the exact nodal values. The **FEM and FVM** provided a close but approximate solution, thereby successfully demonstrating the fundamental trade-offs between different numerical techniques.